MT1011 Engineering Materials

Mechanical properties & Measurement

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Tensile properties

Elastic and plastic deformation

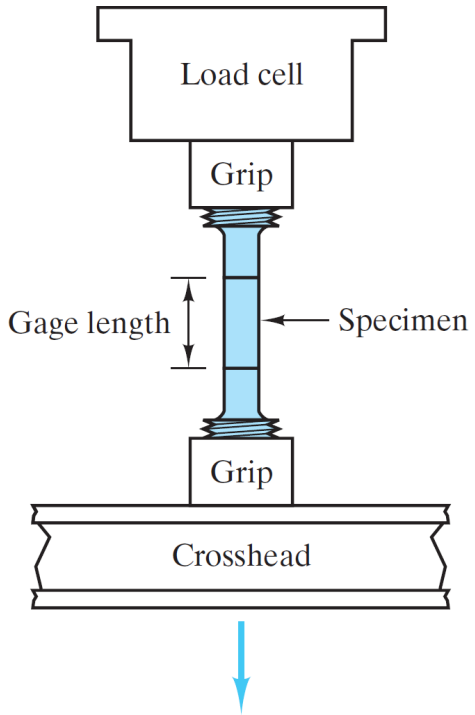
- Most structures are designed to ensure that only elastic deformation occurs when a stress is applied.
- A plastically deformed structure or component will have a permanent change in shape.
- Such parts may not be capable of functioning as intended!



Tensile properties

The tensile test

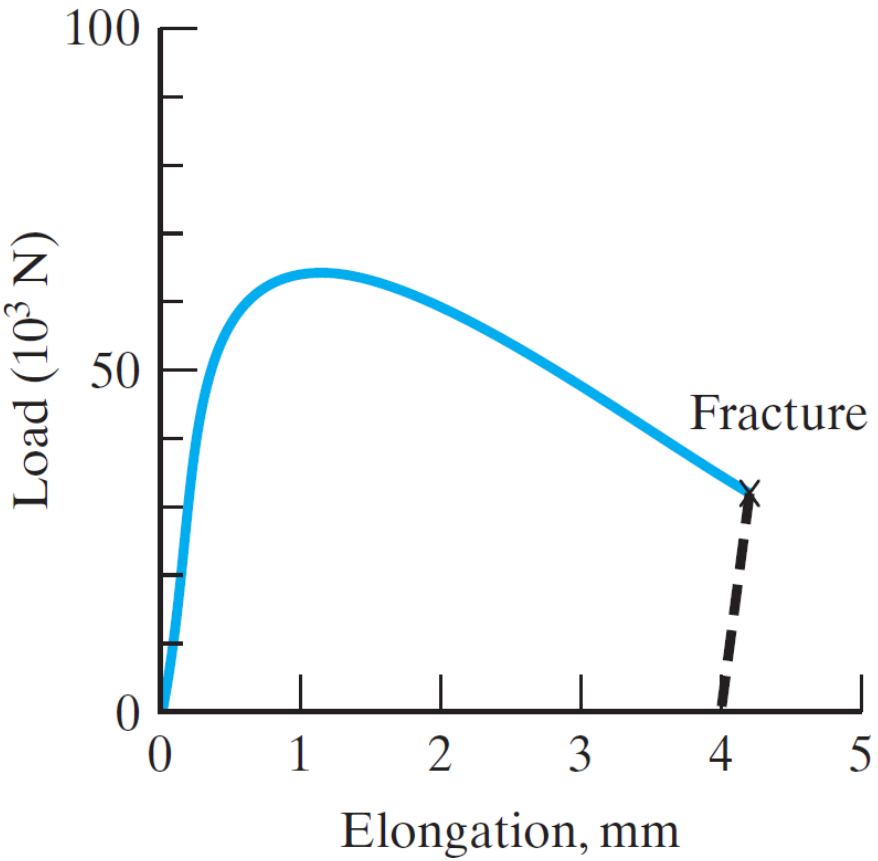
- The simplest questions that a design engineer can ask about a structural material are:
  - How strong is it?
  - How much deformation must I expect given a certain load?
- This basic description of the material is obtained by the tensile test.



Tensile test

Tensile curve

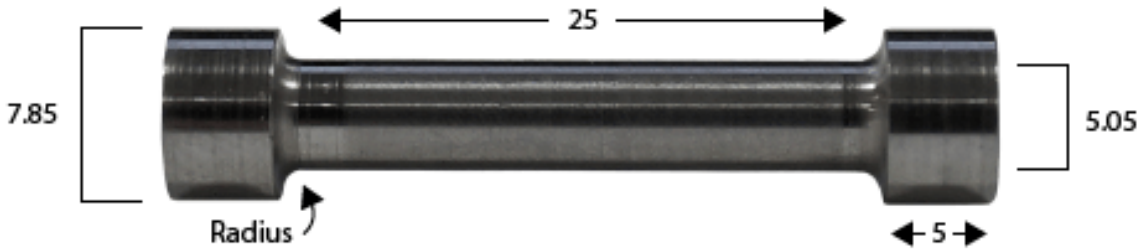
- The load necessary to produce a given elongation is monitored as the specimen is pulled in tension at a constant rate.
- A load-versus-elongation curve is the result of such a test.



- The engineering stress is defined as:

$$\sigma = \frac{F}{A_0}$$

- F: load on the sample with an original (zero-stress) cross-sectional area, A_0.
- Sample cross-section: region near the center of the specimen length.
- The cross-sectional area in this region is uniform and smaller than at the ends gripped by the testing machine.



All dimensions are in millimetres and are approximate

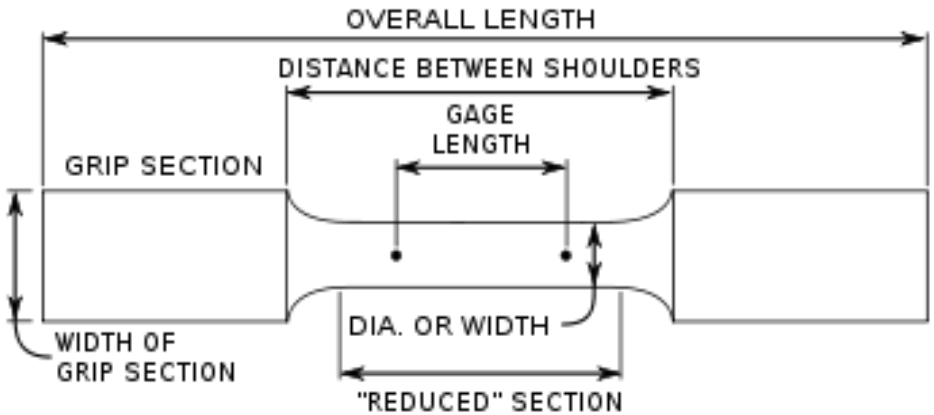
Tensile test

Engineering strain

- This smallest area region guage length experiences the largest stress concentration
- Most deformation at high stress occurs in this region.
- The engineering strain, ϵ, is defined as

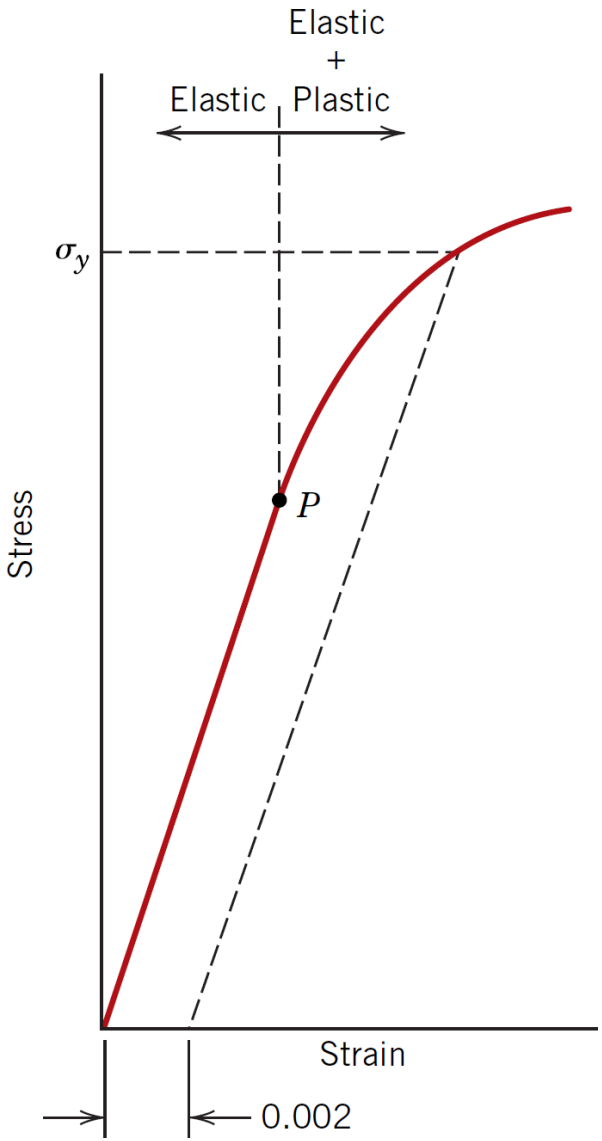
$$\epsilon = \frac{l - l_0}{l_0} = \frac{\Delta l}{l_0}$$

- l: the gauge length at a given load
- l_0: the original (zero-stress) length



Tensile test

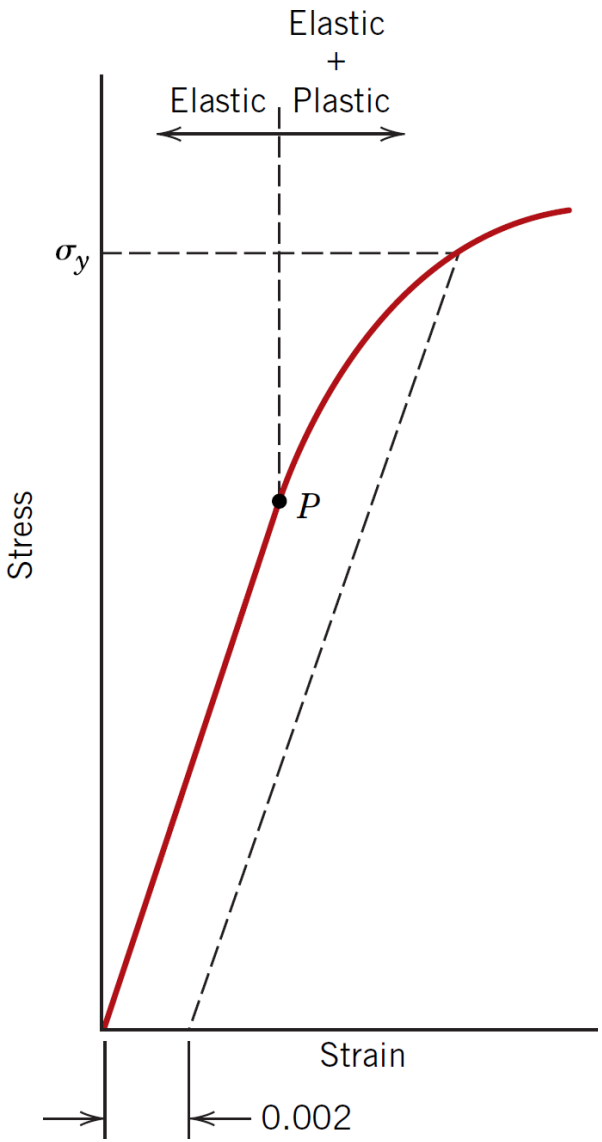
Elastic and plastic deformation



- **Elastic deformation**: Temporary deformation, fully recovered when the load is removed.
- **Elastic region**: Initial linear portion of the stress-strain curve
- **Plastic deformation**: Permanent deformation, not recovered when the load is removed, although a small elastic component is recovered.
- **Plastic region**: Nonlinear portion generated once the total strain exceeds its elastic limit.

Tensile test

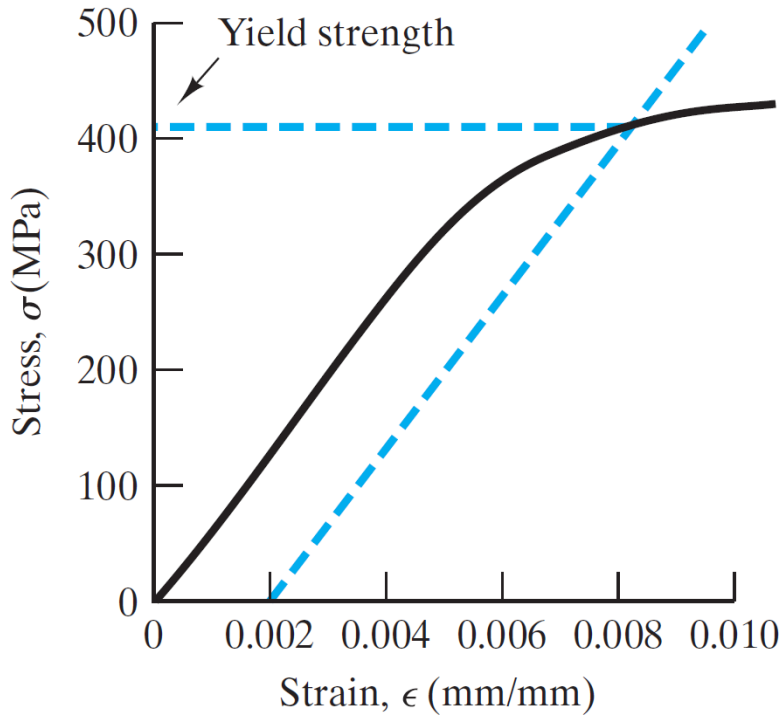
Elastic and plastic deformation



- It is sometimes difficult to find the point where the curve deviates from linearity (straight line behaviour) and enters the plastic region.
- The usual method is to define the yield strength (σ_y) as the intersection of the deformation curve with a straight line parallel to the elastic portion and offset 0.2% (0.002) on the strain axis.

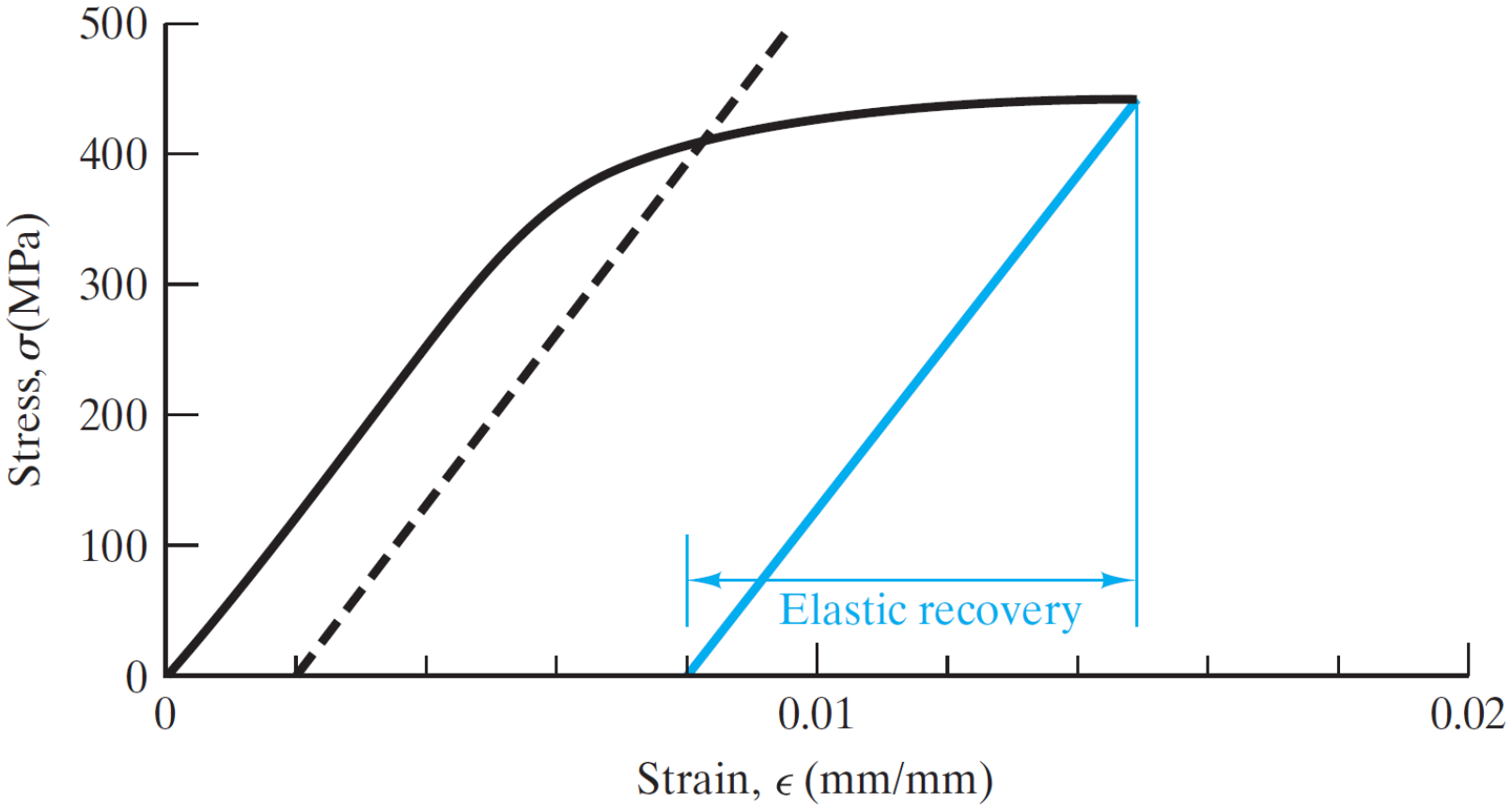
Tensile test

Proof stress



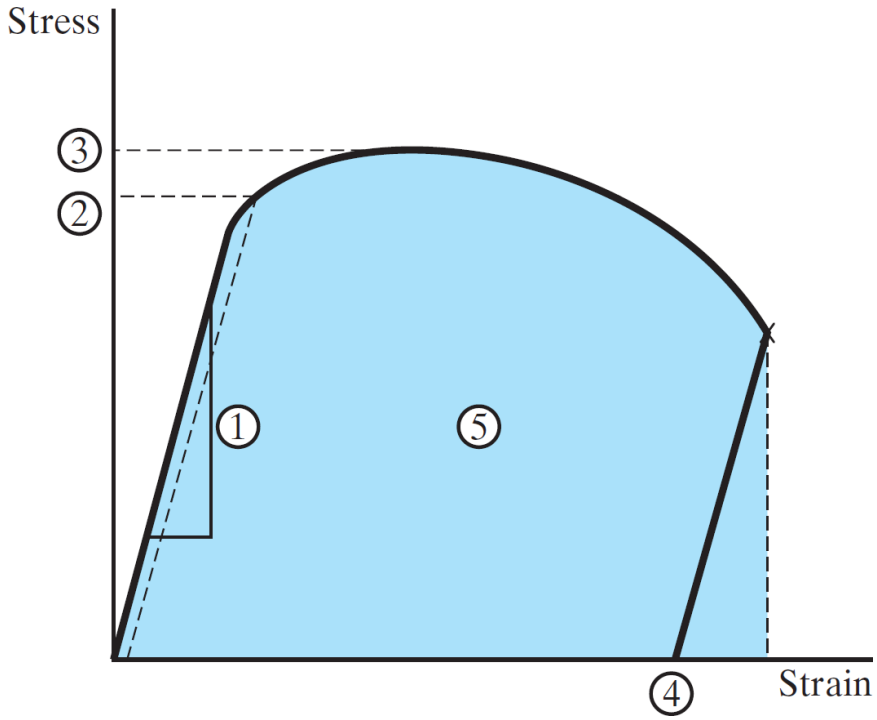
- The yield strength represents the stress necessary to generate this small amount (0.2%) of permanent deformation.
- The dashed line ndicates the small amount of elastic recovery that occurs when a load well into the plastic region is released.

Elastic recovery occurs when stress is removed from a specimen that has already undergone plastic deformation.



Tensile test

Summary of properties obtained from a tensile test



1. Modulus of elasticity, E
2. Yield strength, σ_y
3. Tensile strength
4. Ductility (Plastic deformation)
5. Toughness (Measured under load, dashed line is vertical)

Tensile test

- The slope of the stress-strain curve in the elastic region is the modulus of elasticity, E, also known as Young's modulus.
- The linearity of the stress-strain plot in the elastic region is a graphical statement of **Hooke's law** (?):

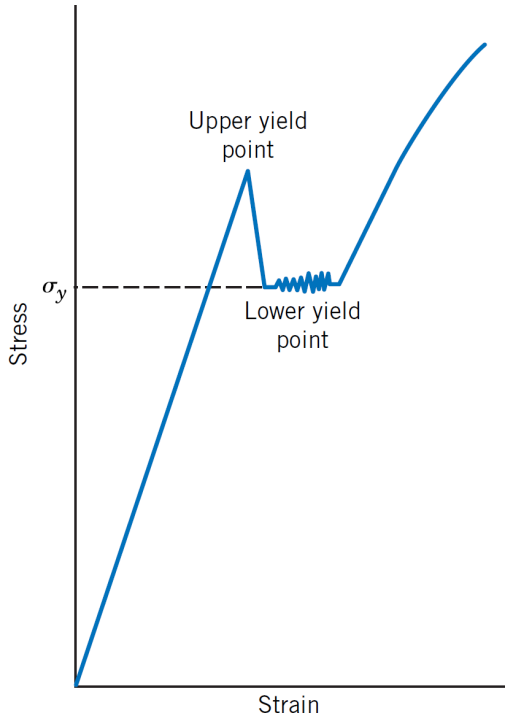
$$\sigma = E\epsilon$$

- The modulus is a **highly practical** piece of information.
- It represents the **stiffness** (resistance to elastic strain).
- It gives the amount of deformation in normal use **below the yield strength**.
- Also the **springiness** of the material during **forming**.



Tensile test

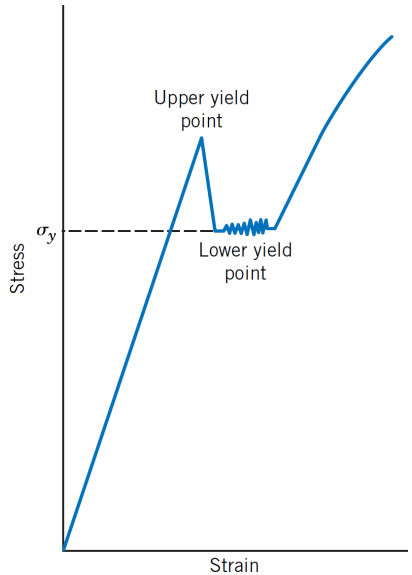
- Some steels and other materials show the tensile stress–strain behavior below.
- The elastic–plastic transition is very well defined and occurs abruptly in what is termed a yield point phenomenon.
- At the upper yield point, plastic deformation is initiated with an apparent decrease in engineering stress.



Tensile test

Lower yield point

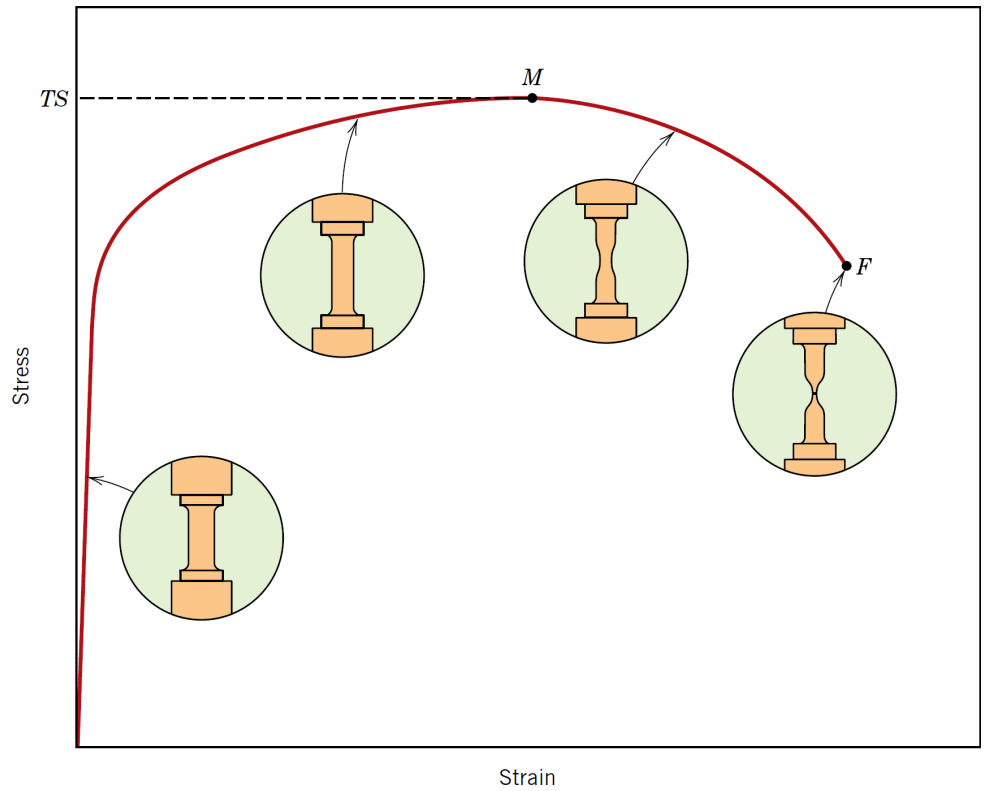
- Continued deformation fluctuates slightly about some constant stress value, termed the **lower yield point**.
- The yield strength is taken as the average stress associated with **the lower yield point**.
- This is because it is **well defined** and relatively **insensitive** to the testing procedure.
- It is not necessary to use the strain offset method.



Tensile test

Tensile strength

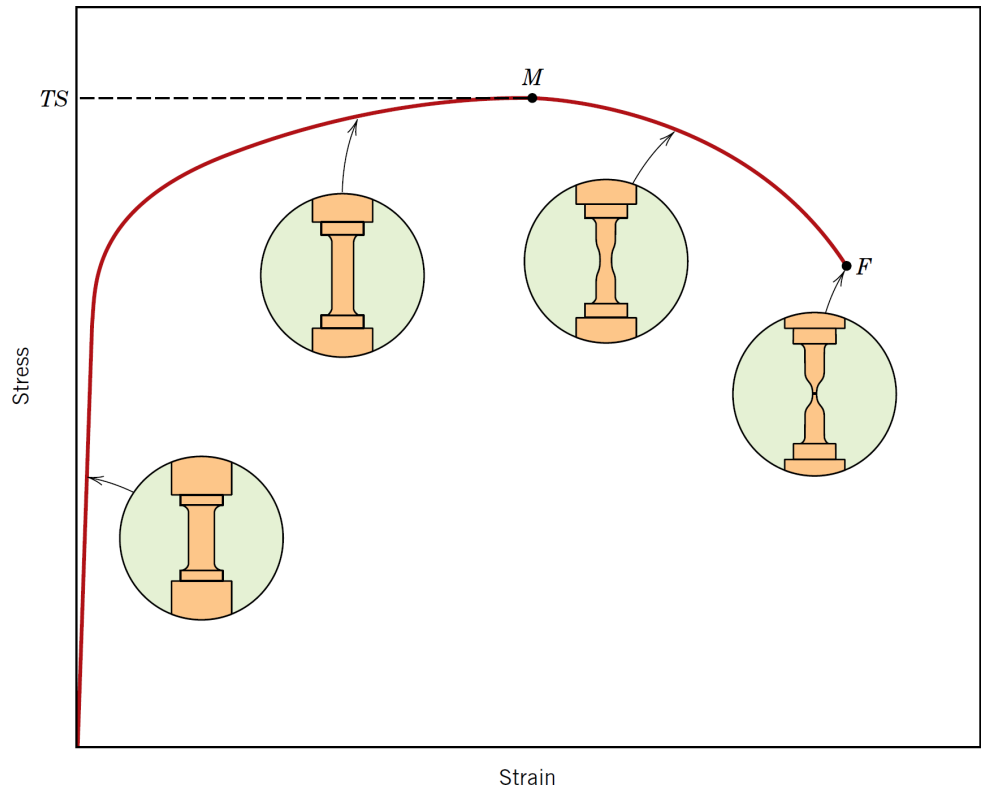
- After yielding, the stress necessary to continue plastic deformation in metals increases to a maximum, point (M).
- After that it decreases until fracture (F)
- The tensile strength TS is the stress at the maximum on the engineering stress–strain curve.



Tensile test

Tensile strength

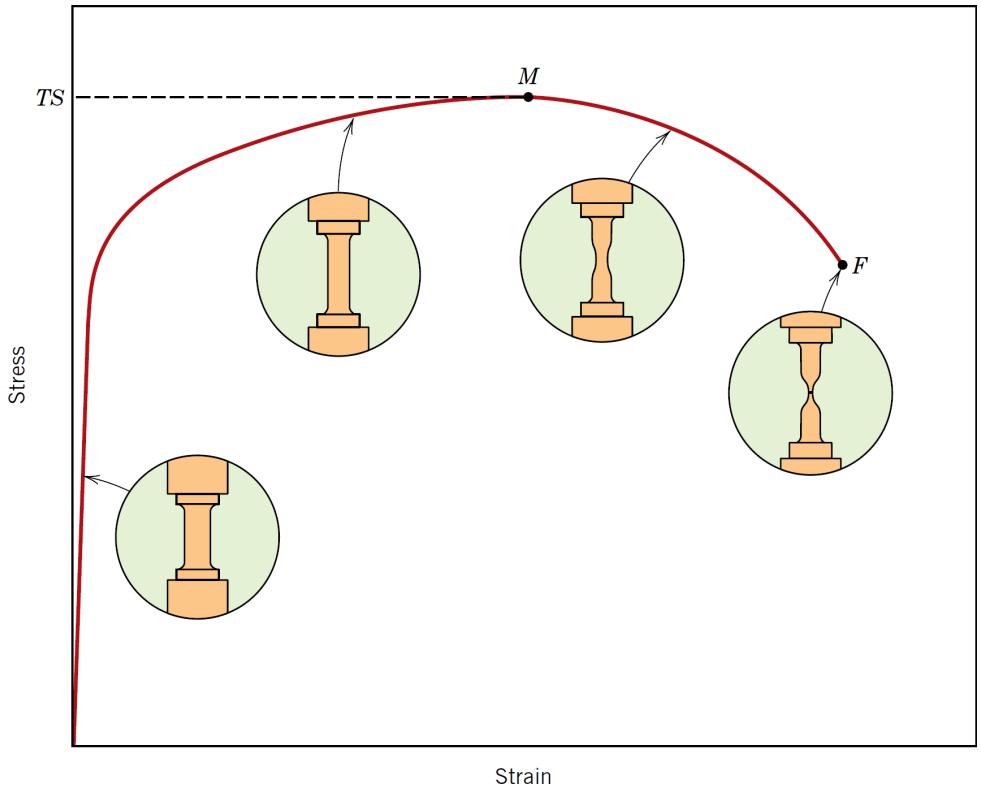
- This corresponds to the maximum stress that can be sustained by a structure in tension.
- If this stress is applied and maintained, fracture will result.
- All deformation to this point is uniform throughout the narrow region of the tensile specimen.



Tensile test

Tensile strength

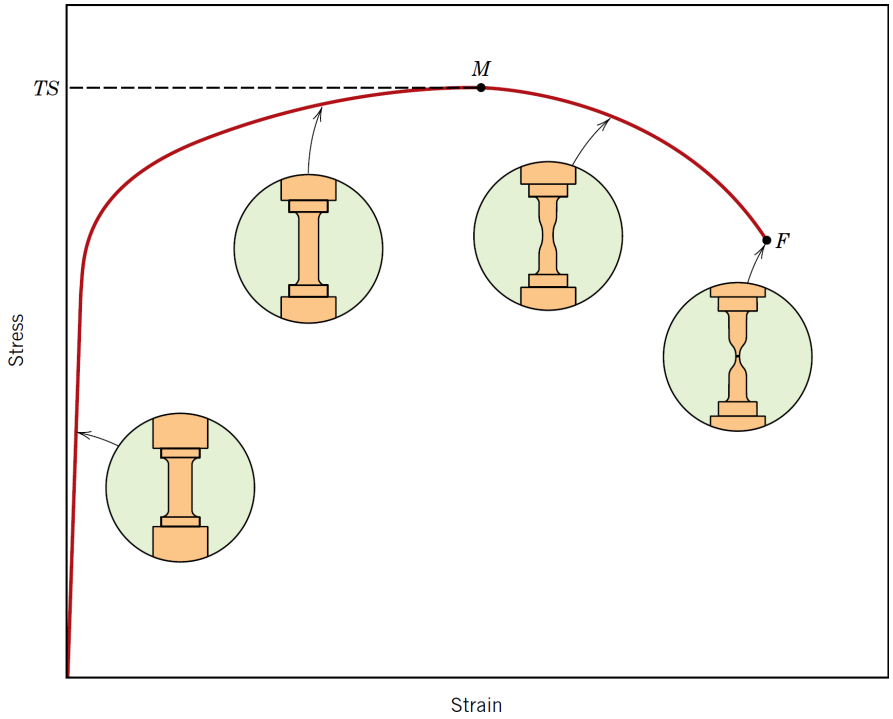
- At the maximum stress, a small neck begins to form.
- After that deformation happens at this neck
- This is known as necking.
- Fracture finally occurs at the neck.
- The fracture strength corresponds to the stress at fracture.





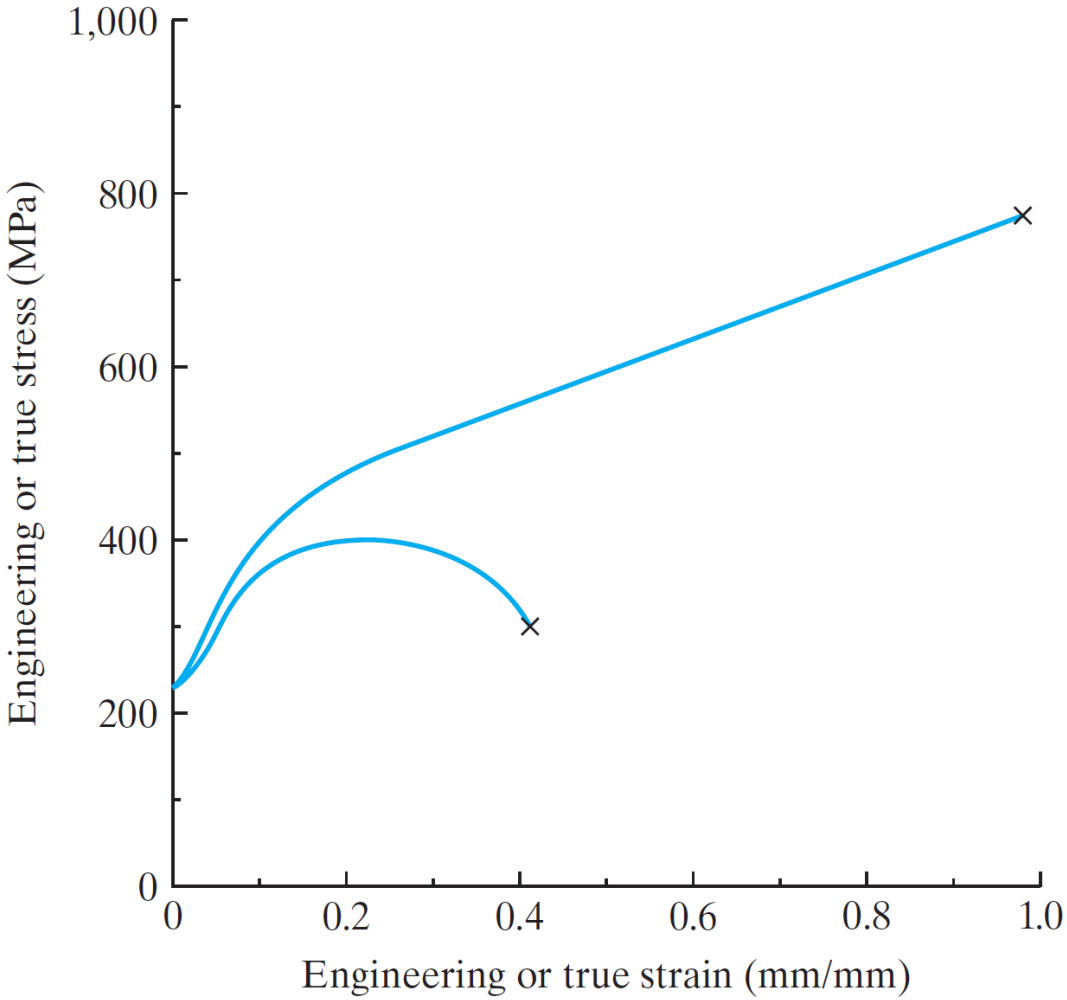
Tensile test

Necking



- Beyond σ_{TS} material appears to soften (stress falls).
- The **engineering stress** and **strain** are defined relative to original sample dimensions (A_0 and l_0).
- At σ_{TS} the sample begins to neck.
- The actual area (A_{actual}) gets smaller and therefore, actual stress increase.

The **true stress** $\sigma_{tr} = F/A_{\text{actual}}$ continues to rise until fracture.



- The engineering stress at failure can be lower than σ_{TS} and even lower than σ_y.
- The σ_f can vary from sample to sample (because of the complexity of the necking process).
- Ductility is given as the percent elongation at failure ($\epsilon_f \times 100\%$).

Percent elongation

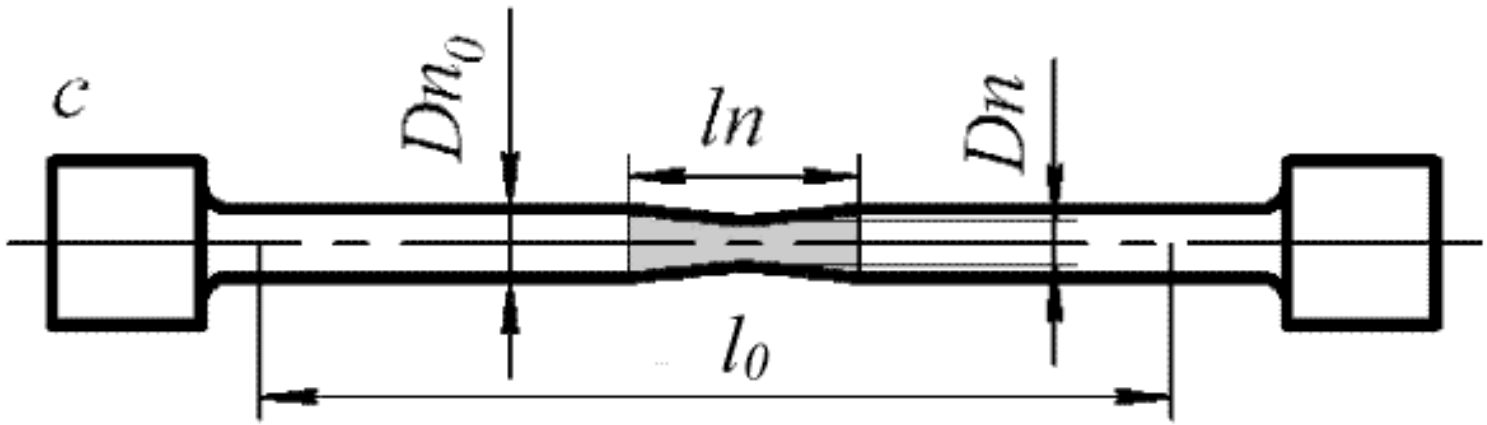
$$\%\text{EL} = \left(\frac{l_f - l_0}{l_0}\right) \times 100\% = \epsilon_f \times 100\%$$

l_f Fracture length

l_0 Original guage length



- A significant faraction (ln/l_0) of the plastic deformation at fracture happens in the neck region.
- Therefore %EL depends on specimen gauge length.
- The shorter the l_0, the greater the fraction of total elongation from the neck.
- Therefore, l_0 should be specified when percent elongation values are cited (commonly $50\,\text{mm}/2$ in.).



Percent reduction in area

$$\%\mathrm{RA} = \left(\frac{A_0 - A_f}{A_0} \right) \times 100\%$$

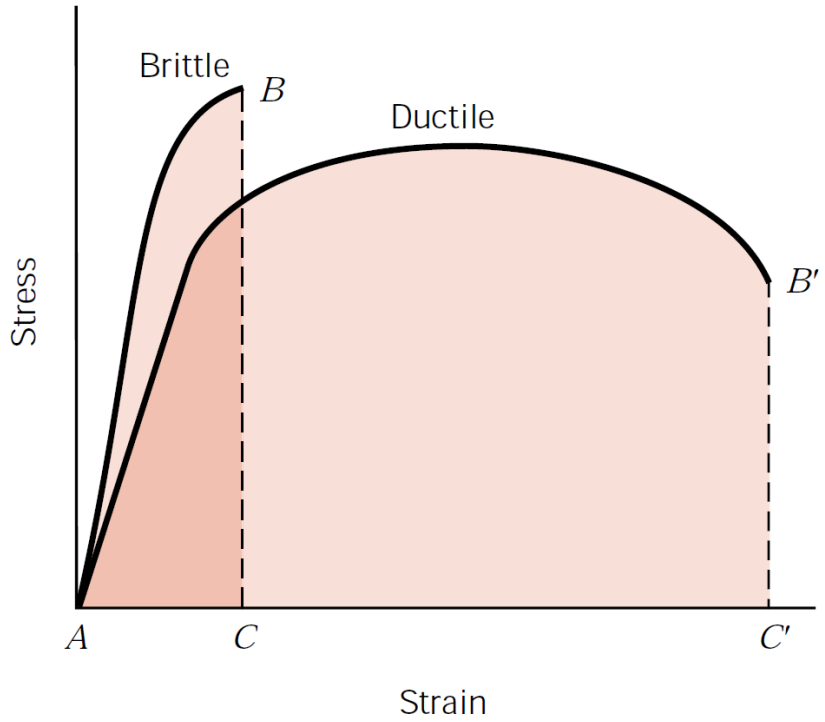
A_0 Original cross-sectional area

A_f Cross-sectional area atfracture



- %RA is independent of both l_0 and A_0 (unlike %EL).
- %EL and %RA are different for a material.

- Most metals have some ductility at room temperature.
- Brittle materials are have a fracture strain ϵ_f of less than about 5%.



Some materials become brittle at low temperatures.

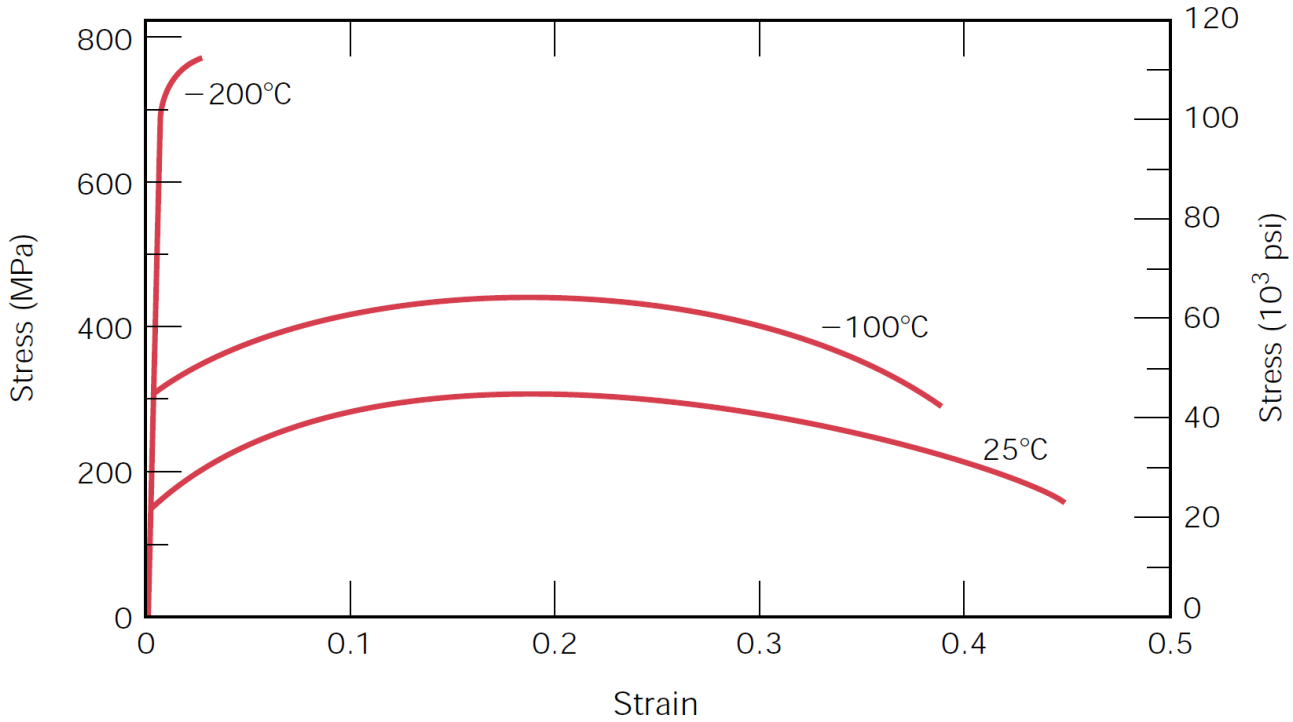


Figure: Stress-strain behaviour of Fe at different temperatures.

Tensile test

Brittle behaviour

- Mechanical properties: also important to ceramics or glasses.
- These material are structurally different to metals.
- Metals normally show some plastic deformation before failure.

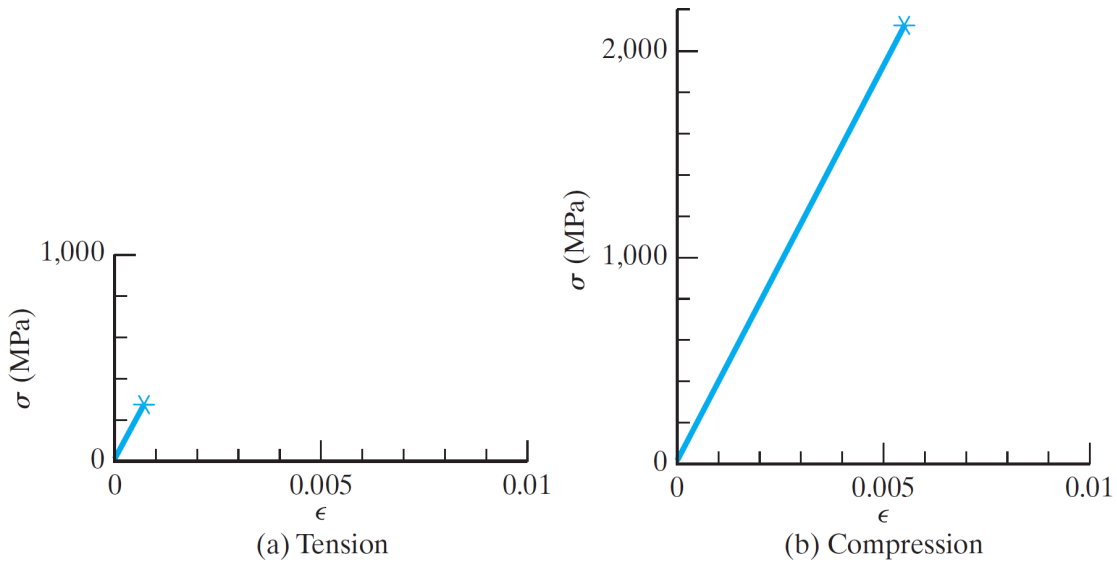


Figure: Dense polycrystalline Al_2O_3 fail at $280\,\text{MPa}$ and $2100\,\text{MPa}$ in tension and compression in the elastic region.

Tensile test

Brittle behaviour

- Ceramics and glasses generally do not show any plasticity.
- Relaively stronger in compression than in tension.
- A beding test is easier (samples easy to produce).
- Failure mode is tensile.

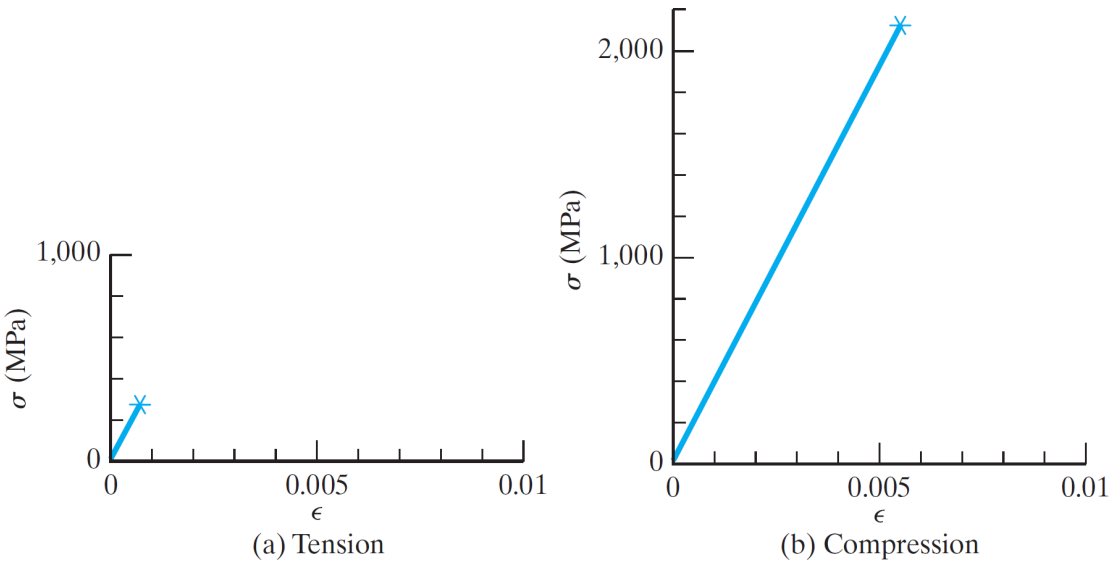
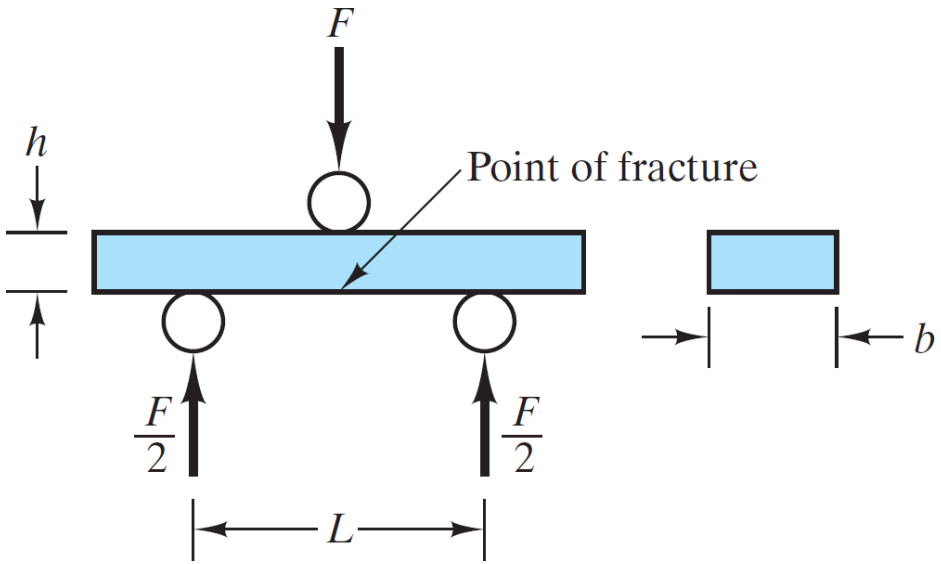


Figure: Dense polycrystalline Al_2O_3 fail at $280\,\text{MPa}$ and $2100\,\text{MPa}$ in tension and compression in the elastic region.

Brittle behaviour

MOR

- The strength of brittle materials is measured as the modulus of rupture (MOR).
- MOR is calculated using a bending test.
- Also called flextural strength.



$$\text{Modulus of rupture} = \text{MOR} = 3FL/(2bh^2)$$

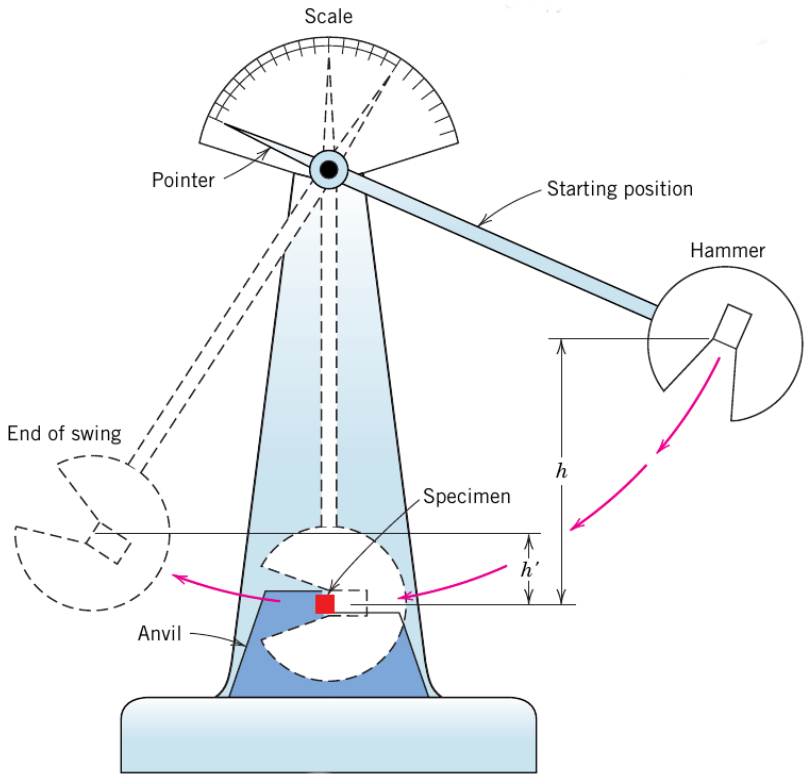
Strength and modulus values of ceramic materials

Modulus of Elasticity and Strength (Modulus of Rupture) for Some Ceramics and Glasses

Material	E (10^3 MPa)	MOR (MPa)
Alumina	380	276–1,034
Sintered[a] alumina (~5% porosity)	370	210–340
Alumina porcelain (90–95% alumina)	370	340
Sintered[a] magnesia (~5% porosity)	210	100
Magnesite (magnesia) brick	170	28
Sintered[a] spinel (magnesia aluminate)	240–260	110–245
Mullite (aluminosilicate)	50–220	150–270
Steatite (magnesia aluminosilicate) porcelain	69	140
Superduty fireclay (aluminosilicate) brick	97	5.2
Sintered[a] beryllia (~5% porosity)	310	140–280
Fully stabilized zirconia	97–207	245
Partially stabilized zirconia	205	600–700
Hot-pressed[b] boron carbide (~5% porosity)	290	340
Silicon carbide	382–476	228–862
Hot-pressed[b] aluminum nitride	345	266
Hot-pressed[b] boron nitride (~5% porosity)	83	48–100
Silicon nitride	120–330	150–1,200
Glass-ceramics	83–138	70–350
Silica glass	73	104
Soda-lime glass	66	150
Borosilicate glass	60	69

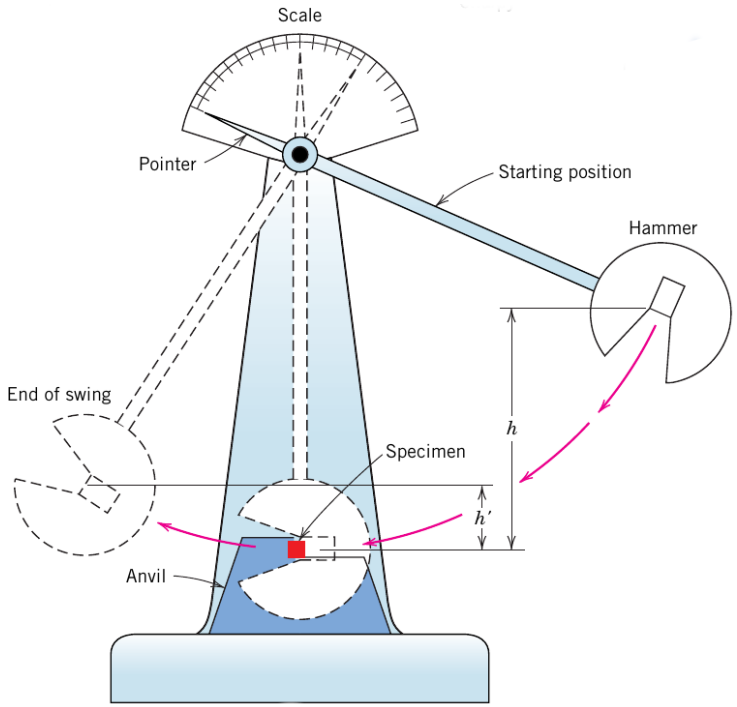
Fracture Toughness

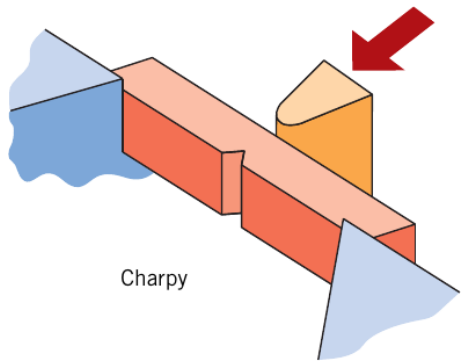
- Impact energy is the energy necessary to fracture a standard test piece under an impact load.
- It is similar to toughness.
- The most common laboratory measurement of impact energy is the Charpy test.



Fracture Toughness

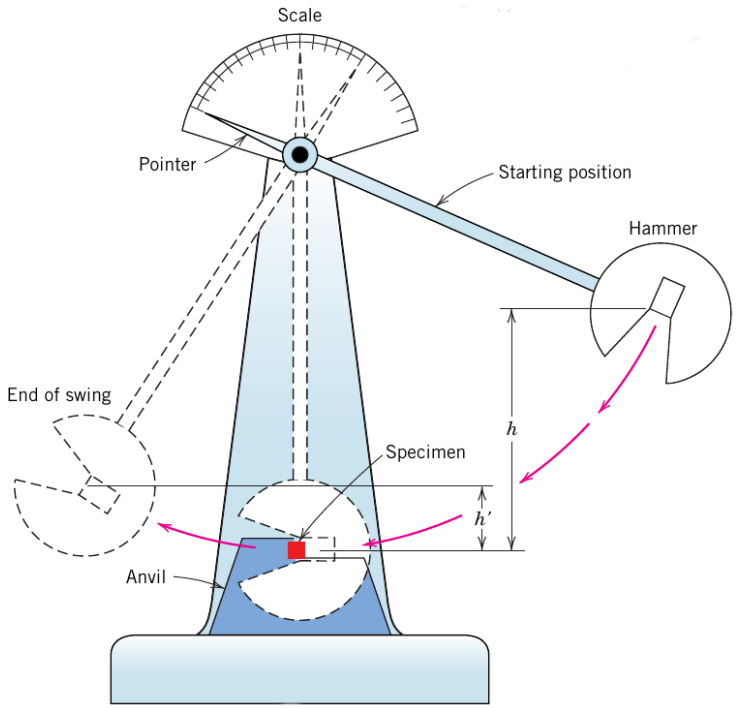
- An impact blow is delivered from a weighted pendulum hammer released from a cocked position at a fixed height h.
- The specimen is positioned at the base as shown.
- On release, a knife edge mounted on the pendulum strikes and fractures the specimen at the notch.
- The notch acts as a point of stress concentration for this high-velocity impact blow.

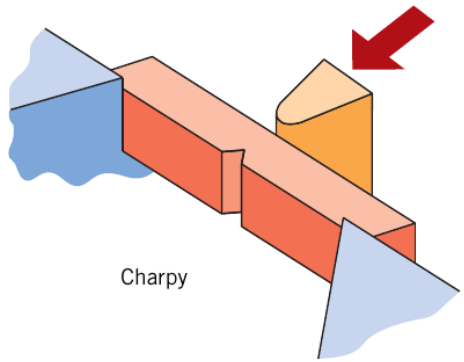




Fracture Toughness

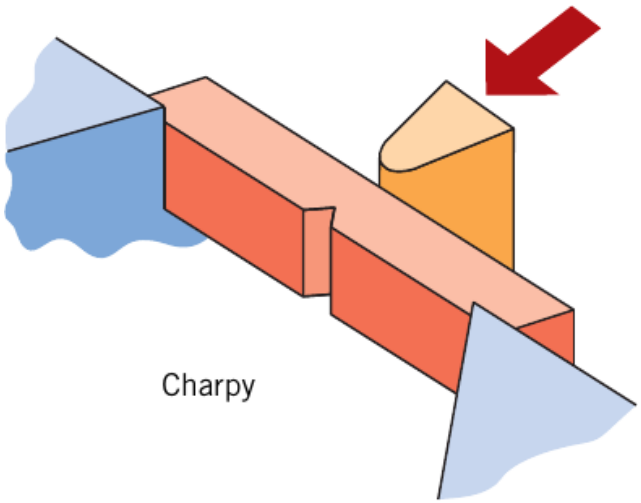
- The pendulum continues its swing, rising to a maximum height h', which is lower than h.
- The energy absorption, computed from the difference between h and h', is a measure of the impact energy.

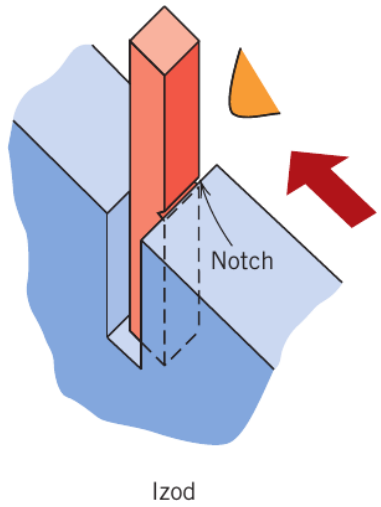




Fracture Toughness

The primary difference between the Charpy and the Izod techniques is the way the specimen is mounted.





Hardness

- Hardness is a measure of a material's resistance to local **plastic deformation** (generally, a small dent or a scratch).
- A small **indenter** is forced into the surface of a material to be tested under controlled conditions of (1) load, and (2) rate of load application.
- The depth or size of the **indentation** is measured and related to a **hardness number**.
- The softer the material, the larger and deeper the indentation, and the lower the hardness index number.
- Measured hardnesses are only relative (rather than absolute).
- Care should be exercised when comparing values determined by different techniques.



Hardness

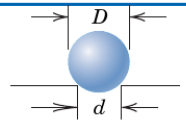
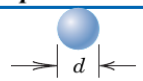
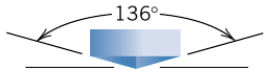
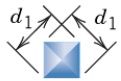
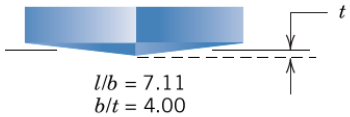
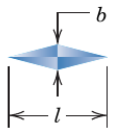
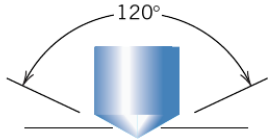
Hardness tests are performed more frequently than any other mechanical test for several reasons:

- They are simple and inexpensive-typically, no special specimen need be prepared, and the testing apparatus is relatively inexpensive.
- The test is nondestructive-the specimen is neither fractured nor excessively deformed -a small indentation is the only deformation.
- Other mechanical properties often may be estimated from hardness data, such as tensile strength.



Hardness

Hardness-Testing Techniques

Test	*Indenter*	*Shape of Indentation* — *Side View*	*Shape of Indentation* — *Top View*	*Load*	*Formula for Hardness Number*[a]
Brinell	10-mm sphere of steel or tungsten carbide			P	$\mathrm{HB} = \dfrac{2P}{\pi D[D - \sqrt{D^2 - d^2}]}$
Vickers microhardness	Diamond pyramid			P	$\mathrm{HV} = 1.854P/d_1^2$
Knoop microhardness	Diamond pyramid			P	$\mathrm{HK} = 14.2P/l^2$
Rockwell and superficial Rockwell	Diamond cone: $\frac{1}{16}$-, $\frac{1}{8}$-, $\frac{1}{4}$-, $\frac{1}{2}$- in. diameter steel spheres			60 kg, 100 kg, 150 kg: Rockwell; 15 kg, 30 kg, 45 kg: Superficial Rockwell	

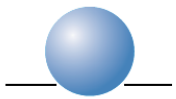


Figure: P in kg, and lengths in mm.

Hardness

Hardness tests are performed more frequently than any other mechanical test for several reasons:

- They are simple and inexpensive-typically, no special specimen need be prepared, and the testing apparatus is relatively inexpensive.
- The test is nondestructive-the specimen is neither fractured nor excessively deformed -a small indentation is the only deformation.
- Other mechanical properties often may be estimated from hardness data, such as tensile strength.



Hardness

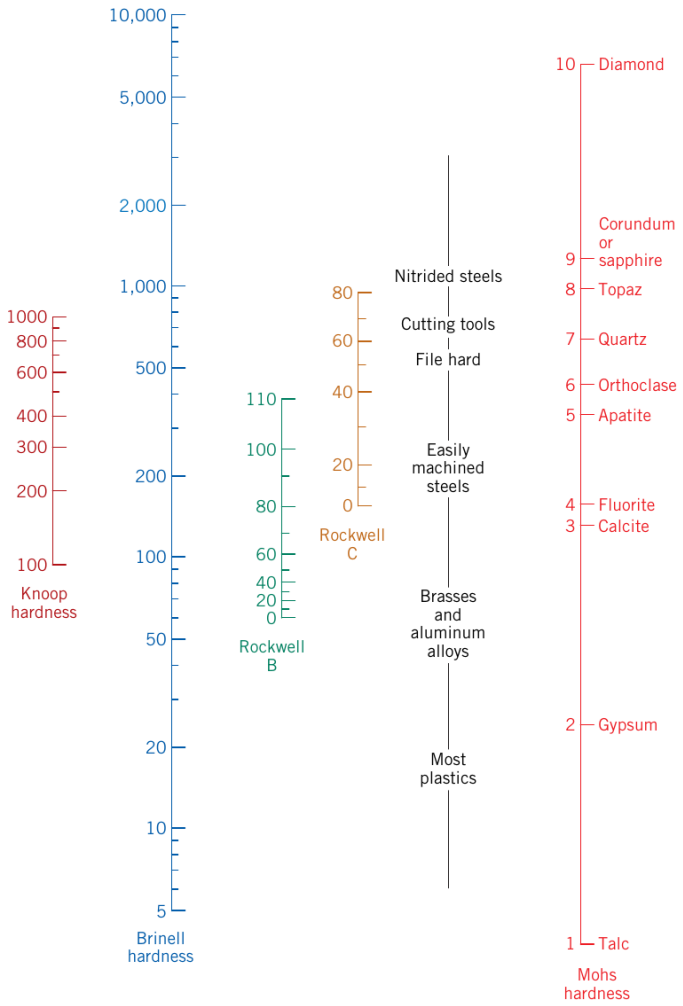


Figure: Comparison of several hardness scales.

